

Week 3: Exchange Rate Expectations, Carry Trades, and Regimes

FINA3020 at CUHK Business School

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Expectations and investment views

- Why currencies jump on unexpected news
- Policy surprises and the expected rate path
- Carry trades, leverage, and spot FX crashes
- Forecasting models and economic insights
- Trading signals, positioning, and model testing

Interpreting markets and policy

- Options volatility across maturities and strikes
- What risk reversals reveal about protection costs
- Exchange rate regimes and policy constraints
- Intervention in several markets
- Evidence that supports or challenges an FX view

UIP connects the interest-rate gap to expected currency adjustment, under risk neutrality:

$$\frac{\mathbb{E}_t[S_{t+\tau}] - S_t}{S_t} \approx (i_H - i_F)\tau$$

i : **annualized** nominal rate. With simple interest over τ years, total interest over the horizon is $i\tau$ (5% p.a. for 3 months gives 1.25%). With compound interest, there is a power or exponential form depending on frequency of compounding.

- **Quote convention:** S tells you how much domestic currency you get for 1 unit of foreign currency (7.8 HKD per 1 USD). Decrease = domestic appreciation.
- **UIP prediction:** the higher-interest-rate currency should depreciate enough [spot] to equalize expected unhedged returns.
- **Evidence from Week 2:** high-yield currencies often fail to depreciate enough at short horizons. This motivates carry strategies, but beware of crash risk.

3 questions for Week 3

Why do spot FX rates jump on monetary news? How do investors interpret parity deviations? How do investors form expectations?

Nominal and real forecast calculation →

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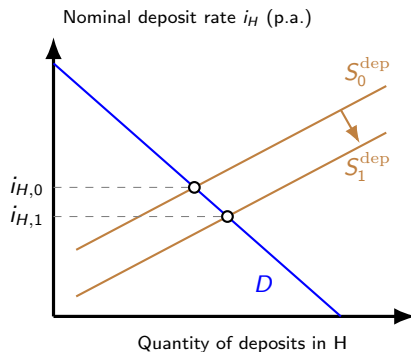
- Let S and F be home currency per foreign unit. With annualized nominal rates and simple interest, CIP:

$$\frac{F}{S} = \frac{1 + i_H \tau}{1 + i_F \tau}, \quad \tau = T - t \text{ in years.}$$

with consistent maturities and day-count conventions.

- A repatriation incentive may increase demand for home currency and home deposits. In a simple deposit-market model, home deposit supply shifts out and its rate falls, holding bank demand fixed.
- If both S and i_H fall while i_F stays fixed, F falls too. Monetary policy and banks' funding choices may offset these responses.

Home-currency deposit market



Brown: deposit supply shifts out.
 Blue: banks' deposit demand is unchanged.
 S^{dep} denotes supply, not the spot rate S .

Rate announcements may convey information differing from investors' expectations. Example:

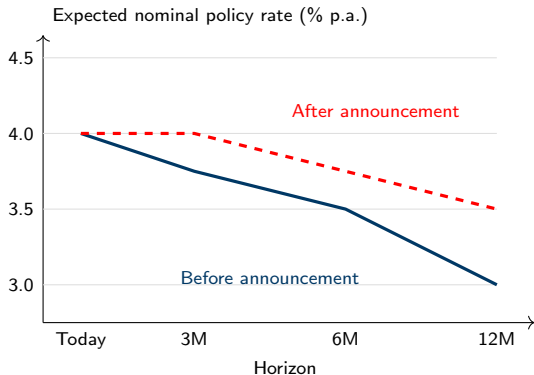
Market expected	Central bank delivers	Interpretation, other news unchanged
25 bp hike	25 bp hike, more hikes signaled	Today's decision was expected. The future path may be more supportive for the currency.
50 bp hike	25 bp hike	Rates rose, but the decision was less hawkish than expected. The currency might weaken.
No change	No change, earlier cuts signaled	Forward guidance can appreciate or depreciate the currency (possibly both at different horizons) even though today's overnight interest rate did not change.

- Compare the expected path in *both* countries.
- A defensive hike during capital flight may accompany currency weakness.

(Policy-path interpretation: Gürkaynak, Sack, and Swanson (2005), IJCB.)

Example: Expected Policy-Rate Path

[Core Concept]



- The current annualized nominal rate stays at 4%. Expected cuts become smaller or later.
- Home currency might appreciate against foreign currency X if X 's path and risk premium stay unchanged.
- Change in interest rates at each horizon. How do we map to a change in spot and forward FX rates?

OIS (overnight indexed swap) and rate-futures prices help infer this path.

(CME FedWatch methodology.)

Historical example: Federal Reserve, 18 September 2024

What the statement said (summary)

The Fed cut its annualized nominal target range by 50 bp to 4.75–5.00%.

Inflation had improved and employment risks had become more relevant.

Further adjustments would depend on incoming data and the balance of risks.

What to interpret

What did the Fed do, how did bond yields respond, compare to what was priced-in.

Identify the reaction function: what could change the next decision?

Which indicators to monitor? The statement did not promise another 50 bp cut at the next meeting.

Market check: compare rate-futures pricing before and after the release, the other country's policy outlook, and the currency's response.

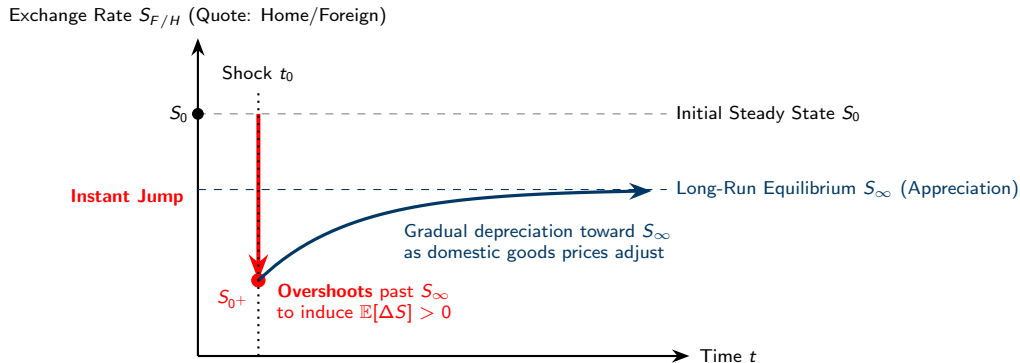
(Federal Reserve, FOMC statement, 18 September 2024.)

Why are exchange rates far more volatile than fundamental macroeconomic variables?

- Key Friction: Goods prices (P) are **sticky** in the short run, while financial asset markets (S, i) clear **instantly**.
- Mechanism: Suppose the domestic central bank unexpectedly raises i_H :
 - ① With prices P fixed, higher i_H contracts money demand and creates an instantaneous domestic return advantage.
 - ② Investors bring in funds ("capital inflow"): the home currency appreciates immediately, so S drops sharply.
 - ③ To restore UIP in financial markets, $(i_H - i_F)\tau \approx \mathbb{E}[\Delta S/S]$, investors must expect the home currency to **depreciate** in the future.
 - ④ The currency must **appreciate beyond** its new long-run equilibrium on impact.

(Dornbusch (1976), Journal of Political Economy)

Mathematical Setup of Dornbusch System →



Initial Baseline: Spot rate $S_0 = 1.3500$ USD per GBP, quoted as *domestic currency (USD) per unit of foreign currency (GBP)*. A **fall** in S is a dollar appreciation. Both one-year nominal rates equal: $i_{\text{USD}} = i_{\text{GBP}} = 3.0\%$.

The Monetary Shock (Time t_0):

- Fed tightening raises the one-year nominal USD deposit rate to 5.0%, a 200 bp spread over GBP in this example.
- Long-Run Equilibrium:** In this stylized calibration, the US price level is assumed to fall by a factor of 1.03/1.05 (−1.905%). With $S = P_{\text{US}}/P_{\text{UK}}$, relative PPP gives a permanently stronger dollar:

$$S_{\infty} = 1.3500 \times \frac{1.03}{1.05} = \mathbf{1.32429} \text{ USD per GBP}$$

- The Friction:** Goods prices are sticky in the short run. Asset markets must clear immediately, so the entire burden of adjustment falls on S .

[Detailed Step-by-Step Overshooting Arithmetic →](#)

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Detailed Step-by-Step Overshooting Arithmetic →

The one-year nominal rates are locked at 5% (USD) and 3% (GBP). The higher USD interest rate must be offset by expected USD depreciation during the one-year adjustment.

The dollar therefore strengthens *beyond* its new long-run level:

$$S_{0+} = S_{\infty} \frac{1.03}{1.05} = 1.3242857 \frac{1.03}{1.05} = \mathbf{1.29906} \text{ USD per GBP}$$

- **Impact:** S falls from 1.3500 to 1.29906, a 3.77% fall in USD per GBP.
- **Long run:** S settles at 1.32429, a 1.905% fall from the initial level.
- **Adjustment:** S rises 1.9417% from its impact level. Sterling bonds then earn 5.0% in USD, matching US cash: zero excess return under the example's assumptions.

The impact move is exactly twice the long-run move *in logs* in the example numbers. Actual # vary.

Detailed arithmetic and model parameters →

Active Prediction Question

Start from equal nominal rates. A Fed surprise raises the one-year USD rate i_{USD} by 100 bp; i_{EUR} stays fixed. Assume both rates are locked for the next year.

Question: Holding expectations about long-run fundamentals fixed, what should happen to:

- ① The **spot rate** $S_{\text{EUR/USD}}$ today (quoted as USD per EUR)?
- ② The **expected future change** $\mathbb{E}_t[\Delta S_{\text{EUR/USD}}]$ over the next year?

Solution:

- **Spot Rate Today:** USD appreciates immediately on the surprise hike $\Rightarrow S_{\text{EUR/USD}}$ **falls** (fewer USD are required to buy 1 EUR).
- **Expected Change:** Because USD overshoots its long-run equilibrium, UIP requires expected USD depreciation from that level $\Rightarrow \mathbb{E}_t[S_{t+1}]$ is **higher** than S_t (EUR expected to rise by $\approx 1\%$ over one year).

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Historically, high-yield currencies have often failed to depreciate enough to offset their interest advantage. The **carry trade** seeks to earn that advantage while taking currency risk.

- ① **Funding Currency:** Borrow in a low-interest-rate currency (examples: CHF at 0.50%, JPY at 1.50%)
- ② **Target Currency:** Convert proceeds at spot S_t and invest in a high-yield money-market asset (examples: USD at 3.75%, MXN at 6.50%, BRL at 14.00%).
- ③ **Unhedged FX Exposure:** Leave spot risk open to collect the interest differential, hope the spot rate does not move against you.

If UIP held, the expected excess return on the carry trade would be zero. If UIP fails systematically, carry traders earn steady positive “excess returns” during calm markets, with large losses during crashes.

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Consider a 1-year carry trade with fixed nominal rates i_H, i_F , funded in currency F (foreign, low yield) and invested in H (home, high yield). Let S_t be units of F per unit of H (example: JPY per USD):

$$\frac{E_1}{E_0} = L \left[(1 + i_H) \frac{S_{t+1}}{S_t} - (1 + i_F) \right] + (1 + i_F), \quad \text{ROI} = \frac{E_1}{E_0} - 1$$

where E_t is equity at time t , $L = \frac{\text{Assets}}{E_0}$ is gross leverage, and debt is $(L - 1)E_0$.

Carry profit & loss (P&L) depends on

- **Interest Spread:** $(i_H - i_F) > 0$. Steady, positive, and predictable in quiet markets.
- **Spot FX Change:** $(S_{t+1} - S_t)/S_t$. Volatile. Appreciation of F against $H \iff S \downarrow$ erodes profits.
- **Reporting Currency:** Returns measured in funding currency F (e.g., JPY), adjust by ΔS for USD-denominated return.
- **Leverage Amplification:** Hedge funds run several times leverage, multiplying small spot moves into massive equity gains or complete liquidation.

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Example: Multi-Scenario Carry P&L

[Core Concept]

Stylized one-year example with all equity, P&L, and returns measured in JPY.

- Equity: JPY15M (US\$100k). Borrow JPY45M at 0.50%, gross leverage is 4×
- Invest US\$400k (JPY60M) at 5.50%. At maturity receive US\$422k, owe JPY45.225M.

Scenario	USD / JPY	Equity (JPY M)	P&L (JPY M)	JPY ROI
Unchanged spot	150	18.075	+3.075	+20.50%
Mild USD depreciation	145	15.965	+0.965	+6.43%
Moderate JPY rally	130	9.635	−5.365	−35.77%
JPY spike	105	−0.915	−15.915	−106.10%

Wipeout: JPY45.225M / USD422k : USD/JPY = **107.17**, a 28.6% fall in USD/JPY, or 40.0% JPY appreciation against USD, from the initial 150. Margin calls can occur before equity reaches zero.

JPY spike could occur in response to sudden rapid inflation + large BoJ hike, and/or a profitability boom for large-cap Japanese firms that boosts Japanese stocks and drives capital inflows.

[Detailed scenario cash flows →](#)

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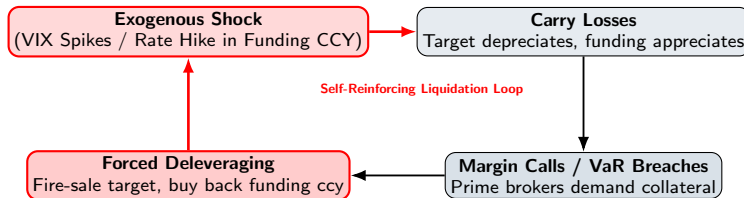
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Carry trades have exposure to sudden currency and funding losses:

- **Negative skewness:** many small gains can be interrupted by large losses. Historical carry returns have substantial left-tail risk.
- **Peso problem:** a rare disaster may not occur in the historical sample. Back-testing without modeling out-of-sample tail risk can overstate ex-ante expected returns.
- **Crash-risk premium:** investors may require compensation for losses in bad states, including market stress and scarce funding liquidity.

(Brunnermeier, Nagel, and Pedersen (2009); Burnside et al. (2011).)

How does a small shock trigger a massive currency (“ccy”) crash? The **Deleveraging Spiral**:



- Crowded trades: Because many funds hold the same long-target & short-funding positions, they all rush to the exit simultaneously, sparking a positive feedback loop.
 - “Positive” refers to sign of amplification multiplier, not good vs bad
- Market liquidity evaporates exactly when investors need it most.

Historical crises show rapid unwinds driven by volatility spikes and funding currency rallies:

Episode	Pair	Trigger & Market Dynamics
October 1998 (LTCM Crisis)	Short JPY / Long USD	Following LTCM collapse and Russian default, funds unwound JPY shorts. JPY appreciated 15% in 48 hours.
October 2008 (GFC Crisis)	Short JPY / Long AUD	Lehman failure triggered global risk-off. "Mrs. Watanabe" and hedge funds liquidated: AUD collapsed >30% against JPY.
August 2024 (Yen Unwind)	Short JPY / Long USD, MXN	BoJ raised policy rate to 0.25% while US data weakened. Leveraged carry unwinds triggered sharp JPY rally and Nikkei drop.

(Aquilina, Lombardi, Schrimpf, and Sushko, BIS Bulletin No. 90 (27 August 2024); Brunnermeier, Nagel, and Pedersen (2009))

Detailed Case Study: August 2024 JPY Carry Unwind →

Week 3 Game: Carry-Trade Tournament

Open Week 3 Game

<https://henryhzhang.com/teaching/fina3020/games/week3/>

- Enter your name, 10-digit CUHK ID, and section.
- Choose a portfolio, respond to the shock, and submit your explanation.
- Keep the tab open until your response is confirmed. Export a copy if it remains unconfirmed.



Scan or click to open

Live FX Game: From Risk to Uncertainty

Open Live FX Game

Scan the QR code, then enter the room code.

- Enter the room code (I will tell you verbally)
- Each round covers one year. Choose an export hedge and a foreign investment allocation.
- Submit each round, observe the common shock and class P&L, then revise.
- First the distribution is known. After a policy shift, the new distribution is not announced.



Scan, then enter the room code

- ① **Stage 1:** Your desk receives \$100,000 USD capital. Annualized nominal rates: USD (3.75%), JPY (1.00%), MXN (6.50%), BRL (14.00%). Borrow JPY; allocate up to 100% across USD, BRL, and MXN. The remainder stays in USD cash at 0%. Choose leverage (1× to 5×).
- ② **Stage 2:** Calculate the one-year ROI if FX is unchanged. Compare with the UIP prediction, then lock the portfolio.
- ③ **Stage 3:** After six months, VIX rises from 14 to 38 and the BoJ hikes by 50 bp. Before the remaining FX move, decide: *sell EM assets and repay JPY, hold, or borrow more?*
- ④ **Stage 4:** Apply the stated stress scenario. Compare USD P&L and debt shortfalls; no interim margin-call rule is modeled.
- ⑤ **Stage 5:** *How did risk budget limits affect your leverage choice during the volatility shock?*

A model can be useful without predicting the exact future spot rate.

- **Point forecast:** How close is the predicted exchange rate to the realization?
- **Relative forecast:** Which currency is likely to outperform another over a stated horizon, including carry?
- **Risk forecast:** How uncertain are you? What could cause a large loss?
- **Investment decision:** Does the expected advantage justify the risk, trading cost, and use of capital?

(Meese and Rogoff (1983); Melvin, Prins, and Shand (2013), Section 2.)

The Melvin–Prins–Shand (2013) Numerical Setup

Universe of 3 currencies: USD (base currency), EUR, and JPY. Current spot quotes: $S_{\text{EUR/USD}} = 1.5000$, $S_{\text{USD/JPY}} = 100.00$.

Trader's Model Forecast:

- Forecasts EUR to appreciate +10% (to 1.6500).
- Forecasts JPY to depreciate −10% (USD/JPY rises to 111.11).
- Portfolio Position: Long +50% EUR, Short −50% JPY (USD neutral).

Out-of-Sample Market Realization:

- EUR *depreciates* by −1.0% (spot ends at 1.4850).
- JPY *depreciates* by −5.0% (USD/JPY ends at 105.26).

(Melvin, Prins, and Shand (2013), Handbook of Economic Forecasting)

USD-based *spot* returns, before carry and costs:

Metric	Random-walk forecast	Trader's forecast
EUR forecast error	−1 percentage point	−11 percentage points
JPY forecast error	−5 percentage points	+5 percentage points
Root mean squared error	3.61%	8.54%
Realized return of stated position	0% (no position)	+2.0%

- The trader lost on the long EUR leg but gained more on the short JPY leg. EUR outperformed JPY by 4 percentage points.
- The ranking of currency returns determines the return on this equal-sized long–short trade, even though the point forecasts were inaccurate.
- To calculate a **Sharpe ratio** $\frac{\text{mean excess return}}{\text{standard deviation of excess return}}$, you need a model of the distribution of outcomes to quantify risk in the denominator

(Teaching illustration of Melvin, Prins, and Shand (2013), Section 2.)

A model turns a set of forecasts into positions that fit a risk budget.

Input	Economic question	Effect on the position
Expected return	How much advantage remains after carry and costs?	A stronger view can support a larger position.
Shared risks	Do several trades depend on the same dollar or risk-sentiment move?	Similar trades may add little diversification.
Risk tolerance	How much loss or volatility can the mandate absorb?	Tighter limits reduce exposure.
Trading costs	Is the revised view different enough to justify trading?	Small forecast changes may warrant no trade.

Example: long AUD and long NZD against USD are separate trades that both depend on the same perspective on regional and global growth.

(Melvin, Prins, and Shand (2013), Section 3.)

Optional mathematical formulation →

Model or signal	Useful insight	Main limitation
Value / PPP	Is the currency expensive relative to a defensible long-run anchor?	A cheap currency can remain cheap. Static valuation metrics like last week's PPP-based may not help inform timing.
Carry	Which currency pays more to hold against the funding currency?	Income can be overwhelmed by depreciation or a funding shock.
Momentum	Has the price or excess-return trend persisted?	A sudden reversal can leave the strategy holding yesterday's winners.
Macro / Policy	Are relative growth and policy expectations changing?	Good news may already be priced or have a different meaning in a new regime.

Different models can disagree because they address different horizons and risks.

(Melvin, Prins, and Shand (2013); Menkhoff et al. (2012, 2017).)

Carry

- Borrow JPY at 1% p.a.; invest in USD at 4% p.a. If spot is unchanged for one year, the spread is 3% before costs.
- JPY appreciation increases the USD cost of repayment. A 10% JPY appreciation can exceed several years of interest income.
- Before increasing leverage, calculate the combined asset loss and debt repayment under a yen rally.

Momentum

- Cross-sectional: rank currencies by past excess returns, then buy winners and sell losers.
- Time-series: buy a currency when its own past excess return is positive; sell when negative.
- Specify the lookback and rebalancing dates. A sudden reversal can hurt both momentum and carry if both hold the same positions.

(Menkhoff et al. (2012), Currency Momentum Strategies; Moskowitz, Ooi, and Pedersen (2012), Time Series Momentum.)

Value models

- Compare the real exchange rate with its history or an estimated equilibrium. With $q = SP_F/P_H$, a high q means home goods are relatively cheap.
- A currency can remain cheap for years. Use valuation to assess a long-run view, not to choose tomorrow's entry price.
- Check whether productivity or terms of trade differ today vs the training sample.

US nominal vs real rates →

Macro models

- Compare new inflation, employment, and growth releases with expectations in both economies.
- Check whether the news changes the expected interest-rate gap. Strong employment matters differently if markets already expected it.
- For a long-JPY view, compare US and Japanese rate-futures repricing. More expected Fed cuts or BoJ hikes can support the view.

(Rogoff (1996); Melvin, Prins, and Shand (2013).)

Real rates and terms of trade →

Real rates across countries →

Example: Conflicting Signals for JPY

[Core Concept]

A fund is considering buying JPY against USD for three months. These are hypothetical observations.

Evidence	Observation	Implication
Valuation	JPY looks cheap relative to a real-value estimate.	Supports a long-run view; gives little information about the next three months.
Policy	Markets price fewer Fed cuts; Japan is unchanged.	The expected US–Japan rate gap widens, working against the JPY position.
Carry	USD yields 4%, JPY 1% p.a.	Roughly 0.75% interest disadvantage over three months, before trading costs.
Positioning	Leveraged funds hold large JPY shorts.	A surprise can trigger short covering. Check changes in gross shorts, not just net positioning.

What would change the decision? A narrowing expected rate gap supports buying JPY. Before entering, calculate the loss if USD/JPY rises 5% and choose the position size that the fund can afford.

Hypothetical example correlations:

	Carry	Momentum	Value
Carry	1.00	-0.15	-0.35
Momentum	-0.15	1.00	-0.40
Value	-0.35	-0.40	1.00

- Calculate correlations from returns measured in the same reporting currency and over the same dates.
- Check the underlying positions. Carry and momentum can both be short JPY before a yen rally.
- Recalculate losses with correlations near +1, wider bid-ask spreads, and less borrowing capacity. Historical diversification may disappear during an unwind.

Compare the combined portfolio with each strategy at the same volatility target, after trading costs.

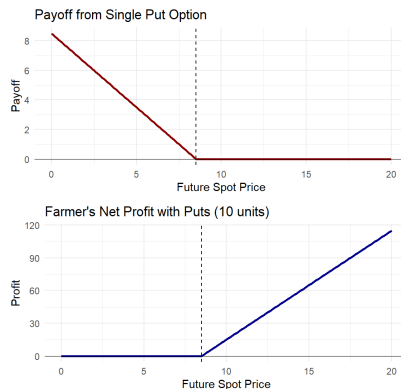
Numerical diversification example →

A **put option** gives its buyer the right to sell at a fixed **strike price**. The buyer pays a **premium** for that right.

Example: A farmer harvests 10 units and owes \$70. A forward locks in \$9/unit. A put has strike \$8.50 and premium \$1.50 per unit.

- **Forward:** deliver at \$9, giving $10(9) - 70 = \$20$ profit at any harvest price.
- **Put:** pay \$15 for a revenue floor of \$85. Minimum profit: $85 - 70 - 15 = \$0$.
- At \$12/unit, let the put expire and sell at market: $120 - 70 - 15 = \$35$ profit.

The option preserves upside, but its premium reduces profit in every outcome. Ignore financing of the premium here.



Let S_T be **JPY per USD** at expiry. Strike $K = 150$ JPY per USD. Each option covers USD 1 and costs $c = p = 3$ JPY today (hypothetical).

- **USD call / JPY put:** right to buy USD 1 for JPY 150. Exercise if $S_T > 150$. Payoff: $\max(S_T - K, 0)$ JPY.
- **USD put / JPY call:** right to sell USD 1 for JPY 150. Exercise if $S_T < 150$. Payoff: $\max(K - S_T, 0)$ JPY.

Expiry spot S_T (JPY per USD)	135	150	165
Call payoff / net profit (JPY)	0 / -3	0 / -3	15 / 12
Put payoff / net profit (JPY)	15 / 12	0 / -3	0 / -3

Payoff versus profit: subtract the premium, ignoring its financing cost. A positive payoff need not cover the premium. Buyer loss is limited to the premium.

Direction: JPY strength means a *lower* S_T , so the USD put provides protection.

Specify the currency, amount, maturity, and reporting currency.

Exposure	Adverse move	Hedge and remaining risk
USD assets financed with JPY debt; report in USD	JPY appreciates: more USD needed to repay the same JPY debt.	Buy the JPY repayment amount forward, or buy a JPY call / USD put. Match the debt maturity and include interest.
USD firm expects EUR receipts in 3 months	EUR depreciates: receipts convert into fewer USD.	Sell confirmed EUR receipts forward; buy a EUR put for uncertain sales. A forward is binding, even if sales fall short.
Leveraged BRL and MXN assets	Both currencies fall while lenders demand collateral.	Reduce positions and debt; hold liquid collateral. A JPY hedge alone does not protect BRL/MXN assets or meet a margin call.

(Hedge directions follow the stated exposures. CME, Managing Currency Risks with Options.)

Example: How Much JPY Risk Is Hedged?

[Core Concept]

A USD-reporting fund owes JPY30M in 3 months, including interest. Spot USD/JPY is 150. It buys JPY15M forward at 149: a 50% hedge. Quotes are JPY per USD.

USD/JPY at repayment	150	135	165
Unhedged: JPY30M / spot	US\$200,000	US\$222,222	US\$181,818
50% hedged: JPY15M / 149 + JPY15M / spot	US\$200,671	US\$211,782	US\$191,580

- At 135, the hedge reduces repayment cost by US\$10,440. At 165, it costs US\$9,762 more than remaining unhedged. A forward removes upside as well as downside on the hedged amount.
- A full hedge fixes repayment at $\text{JPY30M} / 149 = \text{US\$201,342}$. Compare that cost with USD asset proceeds and funding costs before calling the trade profitable.
- Check maturity, notional, collateral terms, and counterparty exposure. A hedge that pays at maturity may still require cash collateral beforehand.

(Hypothetical forward quote and arithmetic. No fees; all costs rounded to the nearest USD.)

Order flow reveals which investors are acting on their information or constraints.

- Signed flow distinguishes buyer-initiated from seller-initiated trades. Every trade has a buyer and seller, but usually one side demands immediate execution (market order).
- Dealers learn from incoming trades and adjust prices as they absorb inventory and infer demand.
- Evans and Lyons explain 64% of daily USD/DEM variation and 46% of USD/JPY variation in their sample when they include interdealer flow.
- Those results use flow from the *same day*. A usable forecast needs information available before the trade and evidence on subsequent returns.

(Evans and Lyons (2002), Table 1. Four-month sample, May–August 1996.)

Participant	Possible reason for the FX trade	What the analyst should check
Asset manager	Buys overseas assets, rebalances a portfolio, or adjusts an existing currency hedge.	Does the FX flow reflect a new asset view or a hedge adjustment?
Leveraged fund	Trades a macro view, follows a trend, or reduces positions to meet a risk limit.	Is the flow building exposure or closing an old position?
Corporate	Converts operating cash flows or hedges future receipts and payments.	Is timing tied to business cash flows rather than a directional forecast?

Coverage matters: one dealer or custodian sees a client sample that might not be representative.

(Menkhoff et al. (2016), Information Flows in Foreign Exchange Markets; State Street, 2026 outlook.)

CFTC Traders in Financial Futures: a public window into positioning

- Separate leveraged funds from asset managers. Different mandates can produce opposite positions in the same currency.
- Read long and short contracts, their net balance, and changes over time. Compare net exposure with open interest or its own history.
- A smaller net short can reflect short covering, new longs, or both. Net positioning is a stock, not a direct measure of cash entering FX.
- Reports usually arrive on Friday with positions from Tuesday. They cover reported futures markets, not all spot and OTC activity.

Example of an interpretation error

A hedge fund's yen-futures short may hedge an OTC position. The participant label alone does not reveal the fund's total currency view.

(CFTC, Traders in Financial Futures explanatory notes; COT release schedule.)

Hypothetical release: employment is stronger than the published consensus.

Market setting	Possible response	Useful evidence
The news changes policy expectations	Expected rates and the currency rise.	Rate-futures repricing supports a policy explanation.
Investors anticipated an even stronger number	The currency stalls or falls despite the consensus beat.	Earlier price moves and positioning suggest an optimistic starting point.
Another shock dominates	The currency falls while global risk sentiment deteriorates.	Cross-market moves and the other currency's news challenge a single-release story.

(Illustrative mechanisms. Policy surprises: Gürkaynak, Sack, and Swanson (2005).)

Suppose a USD-reporting fund sets a US\$10,000 loss limit for a JPY-financed position. Choose the rule before testing it.

- ① Measure the joint loss. Revalue assets and debt under JPY appreciation of 10%, BRL depreciation of 15%, and MXN depreciation of 10%, all against USD. Include accrued interest.
- ② Reduce the exposure if the loss exceeds US\$10,000. Sell enough assets and repay JPY to meet the limit. Buying JPY forward only hedges the debt's FX risk; it leaves losses on BRL and MXN assets.
- ③ Check cash needs. Add collateral calls and the cost of trading at wider bid–ask spreads. Assets that cannot be sold quickly are poor collateral buffers.
- ④ Test the rule. Use data available before each trade. Compare net returns, maximum drawdown, and limit breaches with an unchanged portfolio at similar average exposure.

(Hypothetical risk limit and stress scenario.)

Historical timing experiment →

Research input	Model insight	Analyst's interpretation
Central-bank language	Tone measures summarize hawkish or dovish communication and disagreement.	Has guidance changed relative to previous statements and market pricing?
Frequent price data	Alternative inflation indicators can flag changes before an official release.	Is coverage representative, and would this change the policy outlook?
Relative macro signals	Statistical models combine growth, inflation, and other information across currencies.	Does the relationship survive new data, different periods, and trading costs?

- Use the information available then. A Friday positioning report cannot predict a trade executed on Tuesday. Use original macro releases, not later revisions.
- Separate estimation and evaluation. Estimate on earlier dates, trade on later dates, and roll the window forward. Choose parameters without inspecting the next period's return.
- State the benchmark. For spot forecasts, compare RMSE with an unchanged-spot forecast. For trades, compare with a fixed carry strategy using the same reporting currency, leverage, and trading dates.
- Assess the return distribution. Report net mean excess return, volatility, Sharpe ratio, maximum drawdown, and worst month. Repeat across subperiods; check whether one crisis explains the result.

(Melvin, Prins, and Shand (2013), Sections 2–3.)

- **Strike and moneyness:** a USD call is in the money when spot exceeds its strike. A USD put is in the money when spot is below its strike. At the money means near the current spot or forward, depending on the quote convention.
- **Premium today versus payoff at expiry:** an out-of-the-money option can still cost money because it may pay off before it expires or at expiry, depending on its exercise terms.
- **Implied volatility (IV):** enter the observed premium, spot, strike, maturity, and both interest rates into an option-pricing model. Solve for the volatility that reproduces that premium.
- With those other inputs fixed, higher IV means a higher call or put premium. Across different strikes, *higher IV alone does not establish a higher cash premium.*

Reading the next charts

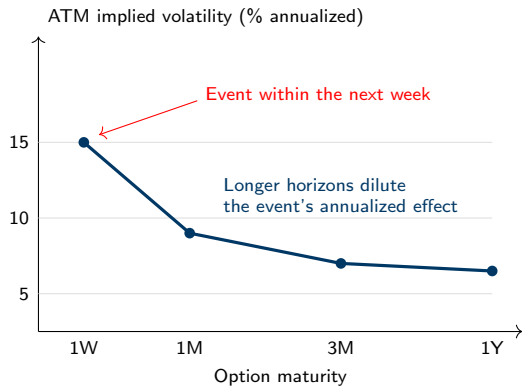
The term structure compares IV across maturities. The smile compares IV across strikes at one maturity. Neither graph shows an option's payoff.

USD/JPY = JPY per USD. A USD call / JPY put benefits from a rise. A USD put / JPY call benefits from a fall.

Exposure	Option hedge	What to compare
USD-reporting borrower owes JPY	Buy JPY call / USD put: protection when USD/JPY falls.	Match JPY debt and maturity; compare maximum USD repayment cost including premium.
JPY-reporting importer owes USD	Buy USD call / JPY put: protection when USD/JPY rises.	Match USD payment and maturity; compare strike, premium, and unhedged cost.

- Implied volatility is backed out from the option price. Compare the same currency direction, maturity, and delta convention across quotes.
- Realized volatility measures past FX variation. A high implied quote can reflect event risk or demand for protection; it is not a forecast of the FX direction.

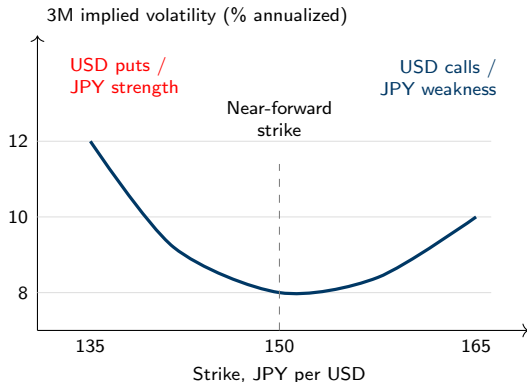
(CME, Managing Currency Risks with Options. OTC currency directions stated explicitly; listed contract conventions can differ.)



- Compare options with similar moneyness, such as at-the-money (ATM), across maturities.
- If payment falls just after an election, compare options expiring before and after the election. Only the latter cover the event.
- 15% annualized volatility over 1W implies roughly $15\% \sqrt{1/52} = 2.1\%$ volatility over that week, not a 15% move.
- Match expiry to the exposure date; a cheap option that expires too early leaves the payment unhedged.

(Illustrative USD/CNH curve, not observed quotes. Event interpretation informed by SGX / Macro Hive, 4 November 2024. Maturity spacing is categorical.)

Hold maturity fixed and compare the volatility quoted at different strikes.



- The wings can sit above the center: the smile.
- A JPY borrower needs the left wing: a USD put / JPY call pays when USD/JPY falls. Here that protection has the higher implied-volatility quote.
- Compare the premium and protected repayment cost at each strike. A cheaper, lower-strike USD put leaves more initial loss with the borrower.

(Illustrative quotes, not market observations. USD/JPY = JPY per USD. CME, FX Options Skews.)

Convention here: USD/JPY risk reversal = USD-call vol minus USD-put vol.

3M options, annualized vol.	Earlier quote	New quote
25-delta USD call volatility	10.0%	10.0%
25-delta USD put volatility	11.0%	13.0%
Call minus put volatility	−1.0 vol point	−3.0 vol points

- A JPY borrower buying USD puts now faces a higher volatility quote. Request premiums for the same strike, notional, and expiry before deciding whether to hedge.
- “25-delta” selects comparably out-of-the-money options by price sensitivity. It is not a literal 25% probability of the outcome.
- Hedging demand, risk aversion, and dealer supply affect the quote. It does not imply a 3% forecast depreciation or a known crash probability.

(Hypothetical quotes. Always check the currency and call-minus-put convention on the data screen.)

Chart 1: USDCNH 1M FX Vol and vol curve inversion at historical extremes

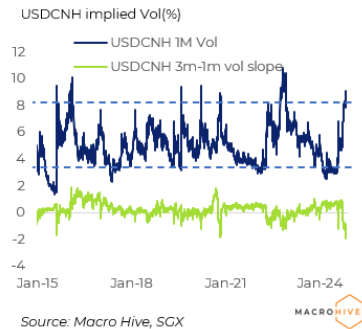
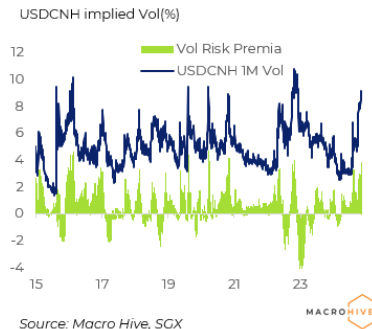


Chart 2: Vol Risk Premia is at 99th Percentile



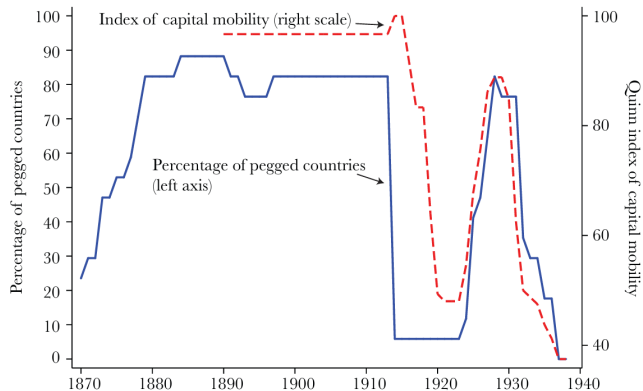
- Left: green is 3M minus 1M implied volatility. A negative reading means an inverted curve.
- Right: the implied-minus-realized measure shows the premium over recent variation.
- The report quoted about 15% annualized 1W volatility across the election: uncertainty, not direction.

(Original historical exhibit: Macro Hive / SGX, Listed FX Options Tracker, 4 November 2024. Panel x-axes are calendar time, not maturity.)

International Monetary Systems: Gold Standard and the Interwar Breakdown

- Classical Gold Standard (1870–1914): Fixed parities and unrestricted two-way convertibility
 - Exchange rates pinned by specie content; cross-border gold arbitrage anchored price levels and payments balance.
 - London served as global liquidity hub: merchant and clearing banks discounted sterling trade bills worldwide.
- Central Bank Policy Dilemma: Tension between external parity and domestic liquidity
 - *Defend parity*: raise discount rates to stem external gold drain (gold earns zero yield).
 - *Emergency liquidity*: under Bagehot's rule, central banks could lend freely at high rates during panics, supported by pre-1914 credibility; this conflict bound far more tightly after 1918.
- War Finance and Interwar Fragility: WWI disrupted convertibility
 - Belligerents suspended convertibility to monetize war deficits, triggering post-war inflation and hyperinflation.
 - Re-pegging at misaligned parities (e.g., UK at overvalued pre-war parity in 1925) generated persistent deflationary pressure.
- Great Depression and Systemic Collapse (1929–1933): “Gold fetters” magnified the crisis
 - Defending gold reserves forced monetary tightening, compounding bank runs, debt-deflation, and credit contraction.
 - Gold standard constrained fiscal deficit finance; capital controls and a flight to gold liquidated global FX reserves.

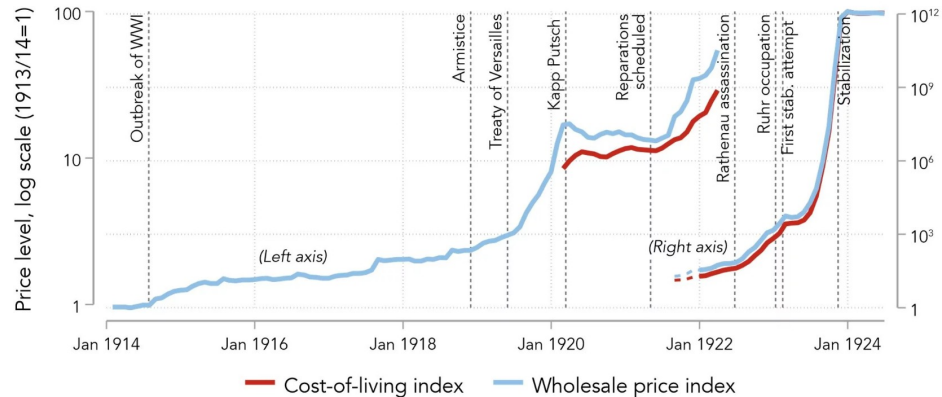
Collapse of the International Gold Standard



Source: Data from Jordà, Schularick, and Taylor (forthcoming) Macrofinancial Database; Quinn, Schindler, and Toyoda (2011).

Historical evidence: capital mobility and the prevalence of pegs changed together across monetary systems.

Semi-Autarkic Interwar Period



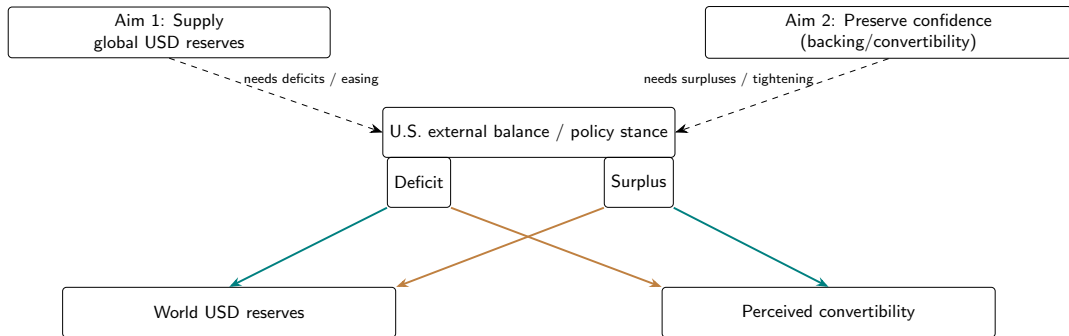
Germany, 1914–24. The log scale shows the size and acceleration of the price-level increase.

Source: Brunnermeier et al (2025).

Bretton Woods and the Original Triffin Dilemma

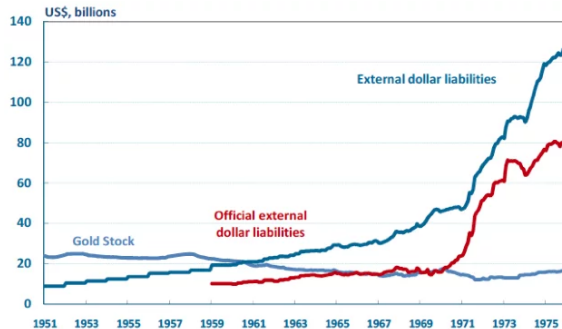
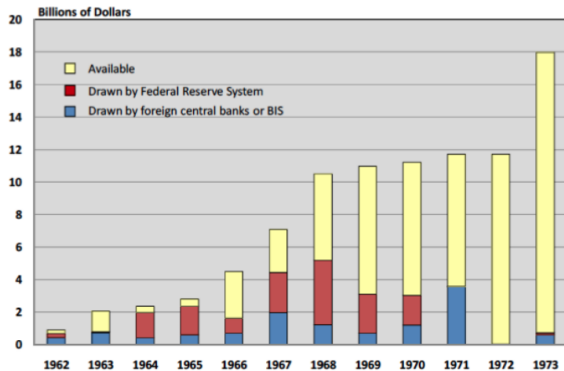
- Bretton Woods Architecture (1944–1971): Fixed exchange rates anchored by the USD
 - USD pegged to gold at \$35 per ounce; member currencies pegged to USD within $\pm 1\%$ parity bands.
 - Capital controls permitted (IMF Article VI) to preserve domestic policy autonomy alongside pegs.
- Reserve Accumulation Mechanics: Balance-of-payments transmission
 - World required USD assets (currency and Treasury securities) for trade invoicing and central bank reserves.
 - Accumulating dollar reserves required net dollar outflows from the US via balance-of-payments deficits.
- The Triffin Paradox (Triffin, 1960): Inherent instability of a national reserve currency
 - Supplying global liquidity demanded persistent US external deficits, expanding foreign dollar liabilities.
 - As foreign dollar holdings eclipsed US gold reserves, credible gold convertibility became mathematically impossible.
- Demise and Collapse (1968–1973): Run on the sovereign gold window
 - Stopgap measures (Federal Reserve swap lines, IMF SDRs) failed to halt private and sovereign gold conversion.
 - US suspended gold convertibility in August 1971 (Nixon Shock); major currencies floated in March 1973.

Bretton Woods System: The Triffin Dilemma Diagram



- US balance-of-payments deficits create global USD assets but erode backing and confidence over time.
- Conversely, US surpluses improve confidence in convertibility but starve the world of liquidity.

Bretton Woods System (BWS) - Demise



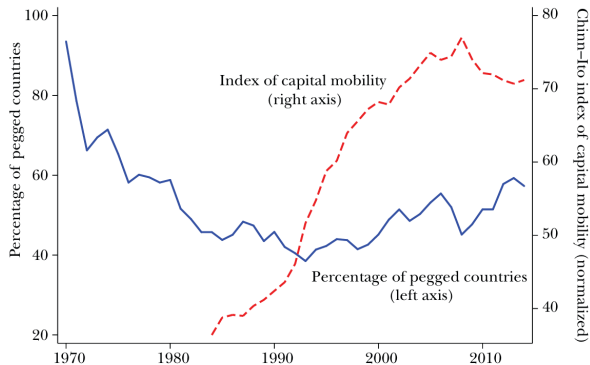
Historical evidence: official dollar liabilities grew relative to US gold, while swap lines provided temporary liquidity.

Source: Michael Bordo; data from Federal Reserve System.

Post-1973 Monetary Arrangements

- Countries chose floating rates, managed floats, pegs, and currency boards. Capital-account liberalization proceeded unevenly.
- A floating currency can still face intervention and monetary spillovers. Financial integration can transmit global funding shocks even without a peg.
- The policy trilemma describes the constraint among exchange-rate stability, capital mobility, and monetary autonomy. Intermediate arrangements involve degrees of each.
- Dollar invoicing and dollar funding persisted after gold convertibility ended. Week 10 connects these uses to demand for safe assets.

Post-1973: Managed Floating Exchange Rates



Notes: Data from Shambaugh (2004) coding and Chinn and Ito (2006) database.

Historical exhibit. Data period, coverage, and units are as labeled.

Exchange-rate arrangements differ in the commitment to a parity and the scope for monetary policy:



- **Hard Pegs (Currency Boards / Unions):** Give up independent monetary policy under the arrangement. Currency boards can still face devaluation or exit risk.
- **Intermediate / Managed:** May target a basket or band, or intervene without announcing a parity. Capital controls are one possible tool, not a requirement.
- **Free Floating:** Let the exchange rate adjust. Monetary policy has more independence, though exchange rates can still affect inflation and financial stability.

Regime map →

Classification over time →

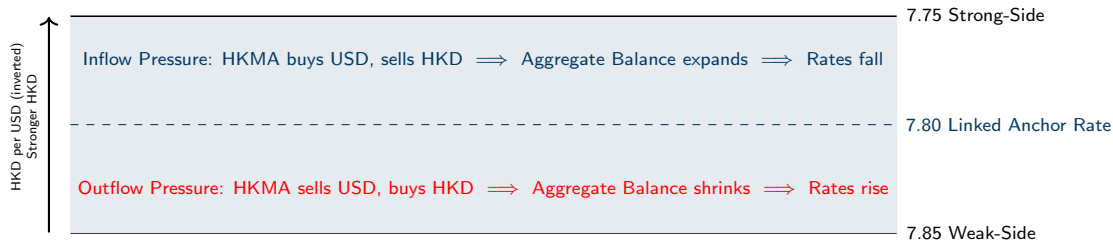
Peg reference currencies →

Currency Union, Dollarization, and a Conventional Peg

- A currency union shares a currency and monetary institutions, as in the euro area. Unilateral dollarization adopts another country's money without a shared policy vote, as in Ecuador.
- A currency board issues domestic monetary liabilities under a rule linking them to foreign reserve backing. Hong Kong retains a separate currency and its own institutions.
- A conventional peg commits the central bank to a rate or band using reserves and policy tools. It need not have currency-board backing rules.
- A crawling peg changes its central rate over time. Managed floating permits rate movements while allowing intervention. De facto behavior can differ from the announced regime.

[Currency use, unions, and pegs map →](#)

Hong Kong's Linked Exchange Rate System (LERS), established in 1983, is a classic Currency Board:



- 100% Foreign Reserve Backing: The monetary base is backed by US dollar assets; commercial bank deposits are separate liabilities.
- Automatic Adjustment Mechanism: When HKD hits 7.85, HKMA sells USD reserves and absorbs HKD, shrinking interbank cash and driving HIBOR upward to stem outflows.

[Detailed HKMA Convertibility Mechanics \$\rightarrow\$](#)

Illustrative trigger: carry outflows push HKD to the weak-side undertaking at 7.8500.
HKMA Intervention (Selling \$1B USD reserves at 7.85):

HKMA Balance Sheet (Exchange Fund)

Assets	Liabilities
FX Reserves (USD): −\$1.0B USD	Aggregate Balance (Bank Reserves): −7.85B HKD

Automatic Monetary Adjustment:

- ① Aggregate Balance: Commercial bank reserves contract (e.g., from 45B to 37.15B HKD).
- ② Interbank liquidity: Reduced reserves can lift HIBOR as liquidity becomes scarce.
- ③ Higher HIBOR: Can narrow the gap against comparable USD rates and reduce the incentive for carry outflows.

Interpretation: The rate response depends on reserve buffers and liquidity demand. A remaining rate gap does not guarantee carry profits.

[Detailed HKMA T-Account Sequence →](#)

The starting liquidity buffer helps explain a persistent interest-rate gap.

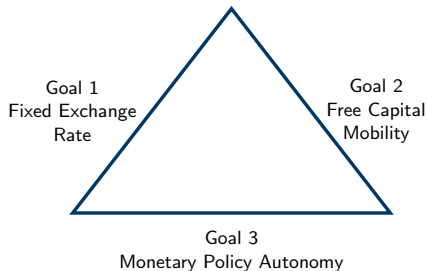
Historical episode	HKMA operation	Market interpretation
Early May 2025	Strong-side undertaking: sold HK\$129.4bn and bought USD.	Added HKD liquidity. Softer HKD rates encouraged subsequent carry outflows.
Late June to mid-August 2025	Weak-side undertaking: bought HK\$119.9bn and sold USD.	Removed much of the earlier injection. The rate response depended on how much liquidity remained.

- At 7.85 HKD per USD, conversion implements the published undertaking. It is not necessarily a discretionary forecast that HKD is undervalued.
- Read the Aggregate Balance alongside comparable-tenor interest rates. An intervention amount alone cannot predict the size of the next HIBOR move.

(HKMA, Half-Yearly Monetary and Financial Stability Report, September 2025.)

A country can simultaneously achieve at most two of the following three desirable goals:

- Goal 1: Fixed Exchange Rate: Limits movement against the anchor currency.
- Goal 2: Free Capital Mobility: Allows cross-border borrowing and investment.
- Goal 3: Monetary Autonomy: Sets domestic policy independently.
- Policy Regimes (Trade-off Corners):
 - *Hard Peg / Currency Board* (Hong Kong, Eurozone): $1 + 2 \implies$ sacrifice 3.
 - *Floating Exchange Rate* (US, Japan, UK): $2 + 3 \implies$ sacrifice 1.
 - *Capital Controls* (China, Bretton Woods): $1 + 3 \implies$ sacrifice 2.
- Global Financial Cycle (Rey 2013): Global financial conditions can limit policy independence even under a float.



(Obstfeld, Shambaugh, and Taylor (2005); Rey (2013))

Trilemma Trade-offs Across Global Economies →

First identify the operation and the objective.

- FX transaction: selling foreign reserves to buy domestic currency adds immediate demand for the domestic currency.
- Liquidity channel: the purchase absorbs domestic bank reserves. If the authority leaves this unsterilized, funding conditions can tighten, depending on the reserve framework.
- Sterilization: an offsetting domestic operation restores bank reserves. This removes that mechanical liquidity contraction.
- Other channels: intervention may signal policy resolve, change the currency risk private investors hold, or supply liquidity in a disrupted market.

The economic distinction

Sterilization describes the offset to bank reserves. It does not by itself establish whether intervention succeeds or whether every market interest rate stays unchanged.

(Sarno and Taylor (2001); BIS Markets Committee, FX interventions (2022).)

Balance-sheet comparison →

Observed headline or data	Reasonable interpretation	What remains unproven
Officials warn against excessive moves.	Verbal intervention may change expectations about future action.	No FX purchase or sale has been confirmed.
Authorities confirm buying domestic currency.	There was an actual transaction in support of the currency.	The size of its causal price impact and its persistence.
Reported FX reserves fall.	Intervention is one possible explanation.	Valuation changes and other reserve transactions must be separated.
The currency rebounds sharply.	Pressure has eased over the observed window.	Intervention may coincide with data surprises, policy news, or short covering.

(Interpretive framework based on BIS Markets Committee, FX interventions (2022).)

Goal: Central Bank sells \$5B USD foreign reserves to defend against currency depreciation.

Central Bank Balance Sheet

Assets	Liabilities
Foreign Reserves (USD): $-\$5B$	Commercial Bank Reserves (M^s): $-\$5B$
Net Change Assets: $-\$5B$	Net Change Liabilities: M^s Falls by $\$5B$

Economic Transmission:

- Domestic banking reserves: Contract by \$5B equivalent.
- Liquidity conditions: Whether reserves become scarce depends on the initial reserve buffer and the operating framework.
- Interbank interest rates: Can rise as reserves become scarce, increasing the funding cost of shorting the domestic currency.

Full Intervention T-Account Comparison →

Operation: Central Bank sells \$5B USD reserves, but neutralizes liquidity impact by simultaneously buying \$5B of domestic treasury bonds in the open market.

Central Bank Balance Sheet

Assets	Liabilities
Foreign Reserves (USD): $-\$5\text{B}$	Comm. Bank Reserves (M^s): $-\$5\text{B}$ (FX Sale)
Domestic Treasury Bonds: $+\$5\text{B}$	Comm. Bank Reserves (M^s): $+\$5\text{B}$ (Bond Purchase)
Net Assets: Bonds $+\$5\text{B}$, FX $-\$5\text{B}$	Net Liabilities: M^s Unchanged (0)

The Limits of Sterilization

The offset restores bank reserves. Signaling and changes in private risk holdings can still affect market prices. Persistent defense faces limits when intervention conflicts with monetary policy or uses scarce foreign reserves.

[Full Intervention T-Account Comparison →](#)

Historical case: yen support in July 2024

Date	Confirmed event	What it tells the analyst
11 July	Japan bought JPY and sold USD, worth JPY3,167.8bn.	An actual currency-support operation, later disclosed by the Ministry of Finance.
12 July	Another JPY2,367.0bn of yen purchases.	The two operations totaled JPY5,534.8bn.
31 July	The BoJ raised its policy-rate guideline to around 0.25%.	A separate monetary-policy decision with implications for the expected rate path.

Institutional distinction: the Ministry of Finance decides intervention. The BoJ executes as its agent. A yen rebound may also reflect US news and position unwinds, so the price chart alone cannot isolate the intervention effect.

(Japan MOF, July–September 2024 operations; BoJ, intervention operations; BoJ, 31 July 2024 decision.)

Chart 3: PBoC Managing spot very close to fixing



Chart 4: Risk-reversal skew remains sanguine over the medium-term outlook



- CNY is onshore; CNH is offshore. Both quotes are renminbi per USD; lower means a stronger renminbi.
- Left: compare spot, fixing, and band. A strong fixing may signal resistance to depreciation without proving a reserve sale.
- Right: skew differs by horizon. Offshore pricing also reflects funding, flows, and market segmentation.

(Original historical exhibit: Macro Hive / SGX, 4 November 2024. Panel titles reflect the author's interpretation at that date.)

Record the intervention time, amount, and objective; compare outcomes before and after.

Objective	Measure	Check before concluding
Restore orderly trading	Bid–ask spread, executable depth, and price differences across venues.	Did execution improve for a fixed trade size? Appreciation is not necessary for success.
Slow depreciation	Spot returns and realized volatility over several windows; implied skew at the same tenor.	Was the rebound sustained? Compare the dollar, peer currencies, and policy-rate news.
Maintain a peg or band	Spot relative to the band; reserve sales; overnight and term funding spreads.	Has pressure moved into funding markets? Compare reserve use with available reserves and upcoming obligations.

A before/after price move alone is not a causal estimate. The authority often intervenes precisely when pressure is greatest.

(BIS Markets Committee, FX interventions (2022); teaching applications.)

Topic	What to be able to do
Expectations	Explain why a currency may fall after a rate hike: compare the decision and guidance with prior expectations.
Carry	Calculate interest, asset values, and debt repayment in one reporting currency; stress target depreciation and funding appreciation together.
Forecasting	Compare forecasts with a benchmark using information available at the time. Assess net returns and drawdowns, not only forecast errors.
Options	Choose the currency direction, strike, notional, and maturity that cover the exposure. Include the premium and collateral terms.
Intervention	Separate reserve transactions from policy signals. Assess spreads, depth, exchange rates, and funding conditions against the stated objective.

Looking Ahead to Week 4: Measuring and Hedging FX Exposure

Today we analyzed exchange rates from the macro and trading desk perspectives. Next week we move inside the corporation to examine how firms manage currency risk:

- Measuring Corporate FX Exposure:
 - Transaction Exposure: Contractual cash flows in foreign currencies.
 - Operating / Economic Exposure: Long-term competitive impact of real exchange rate movements.
 - Translation Exposure: Accounting consolidation of foreign subsidiaries.
- The Hedging Toolkit:
 - Forward Contracts vs. Money Market Hedges vs. Currency Options.
 - Evaluate an uncertain foreign receivable and choose the optimal corporate hedge.

Reminder: Problem Set 1 (Weeks 1–2) is due this Wednesday at noon on GradeScope!

Appendix: Dornbusch Overshooting Dynamic Equations

The Full Dornbusch (1976) Four-Equation System: Let m, p, s, y denote logs; time is in years. i, i^* are annualized nominal rates, π^e expected inflation, and $r = i - \pi^e$ the real rate in continuous time.

- ① Money Market Equilibrium:

$$m - p = -\eta i + \phi y$$

- ② Uncovered Interest Parity:

$$i = i^* + \mathbb{E}[\dot{s}]$$

- ③ Goods Market Aggregate Demand:

$$y^d = \bar{y} + \delta(s - p) - \sigma r$$

- ④ Sticky Price Adjustment Equation:

$$\dot{p} = \kappa(y^d - \bar{y}), \quad \kappa > 0$$

Note: The main slide and Example 1 analyze an unexpected monetary contraction ($\Delta m < 0$), where i jumps above i^* , requiring an immediate overshooting appreciation $s_{0+} < s_\infty$; here we present the canonical expansionary system ($\Delta m > 0$), its exact mirror image: with p fixed on impact ($\Delta p = 0$), $m - p$ rises $\implies i < i^* \implies \mathbb{E}[\dot{s}] < 0 \implies s_{0+} > s_\infty$.

Appendix: Example 1 Detailed Overshooting Math

Step-by-Step Breakdown (one-year nominal deposit rates, fixed over the adjustment):

- 1 Initial Conditions: $S_0 = 1.3500$ USD per GBP (domestic per foreign), $i_{\text{USD}} = i_{\text{GBP}} = 3.0\%$.
- 2 Long-Run Target: The tightening is calibrated to lower the long-run US price level by a factor of $1.03/1.05$ (-1.9048%). Since $S = P_{\text{US}}/P_{\text{UK}}$, relative PPP gives $S_{\infty} = 1.3500 \times \frac{1.03}{1.05} = 1.32429$.
- 3 Short-Run Friction: $\dot{p} \approx 0$ on impact. The interest spread is $5.0\% - 3.0\% = +200$ bps in favour of USD.
- 4 Impact Jump Condition: UIP over the 1-year adjustment period requires:

$$\frac{S_{\infty}}{S_{0+}} = \frac{1.05}{1.03} \implies S_{0+} = 1.324285714 \times \frac{1.03}{1.05} = 1.3500 \times \left(\frac{1.03}{1.05}\right)^2 = 1.29906$$

- 5 The Overshoot: In logs, $\ln(S_{0+}/S_0) = 2 \ln(1.03/1.05) = -3.8463\%$ and $\ln(S_{\infty}/S_0) = \ln(1.03/1.05) = -1.9231\%$. The impact move is exactly double. (In simple percentage changes the two are -3.77% and -1.90% ; the difference is pure compounding.)
- 6 Reversion: Over the year, spot rises from 1.29906 back to 1.32429 ($+1.9417\%$). A sterling bond yields $\frac{1.32429}{1.29906} \times 1.03 = 1.0500$ ($+5.0\%$ USD return), earning zero excess return over domestic US investment.
- 7 Caveat: The exact doubling follows from choosing the price-level drop equal to the interest factor ratio ($1.03/1.05$). In continuous time, the impact move is $1 + \frac{1}{\theta\eta}$ times the long-run move, where θ is the exchange rate's equilibrium convergence speed ($\dot{s} = \theta(\bar{s} - s)$) and η is the interest semi-elasticity of money demand (here $\theta\eta = 1$). Reaching equilibrium after exactly one year is a teaching simplification; continuous adjustment is asymptotic.

Appendix: Carry Trade Cash Flows & Payoff Math

Payoff Derivation for a One-Year Leveraged Carry Position: E_0 : initial equity; $L = \text{Assets}/E_0$: gross leverage. Cash flows are in funding currency F ; i_H, i_F are fixed one-year nominal rates:

- ① Borrowing Leg: Borrow $(L - 1)E_0$ in funding currency F at interest rate i_F . Repayment obligation at t_1 :

$$\text{Debt Repayment}_1 = (L - 1)E_0(1 + i_F)$$

- ② Investment Leg: Total assets $L \cdot E_0$ converted to target currency H at spot S_0 (units of F per H). Invested at i_H . Proceeds at t_1 converted back at spot S_1 :

$$\text{Asset Value}_1 = L \cdot E_0(1 + i_H) \cdot \frac{S_1}{S_0}$$

- ③ Net Equity at t_1 :

$$E_1 = E_0 \cdot \left[L(1 + i_H) \frac{S_1}{S_0} - (L - 1)(1 + i_F) \right]$$

- ④ Return on Equity (ROI):

$$\text{ROI} = \frac{E_1 - E_0}{E_0} = L \left[(1 + i_H) \frac{S_1}{S_0} - (1 + i_F) \right] + i_F$$

Appendix: Example 2 Detailed Scenario Carry Trade Ledger

① Initial Balance Sheet (JPY; fixed one-year nominal rates):

- Equity: $\$100,000 \text{ USD} = 15,000,000 \text{ JPY}$ at $S_0 = 150.00$.
- Borrowing: $45,000,000 \text{ JPY}$ at $i_{\text{JPY}} = 0.50\%$ ($L = 4 \times$ gross exposure).
- Total Assets: $60,000,000 \text{ JPY} \div 150.00 = \$400,000 \text{ USD}$ invested at $i_{\text{USD}} = 5.50\%$.

② Maturity Values (1 Year):

- Debt Obligation: $45,000,000 \times 1.005 = 45,225,000 \text{ JPY}$.
- USD Proceeds: $\$400,000 \times 1.055 = \$422,000 \text{ USD}$.

③ Scenario Calculations:

- Scenario 1 ($S_1 = 150.00$): Assets = $\$422\text{k} \times 150 = 63.30\text{M JPY}$. Equity = $63.30\text{M} - 45.225\text{M} = 18.075\text{M JPY}$. P&L = $+3.075\text{M JPY} \Rightarrow \text{ROI} = +20.50\%$.
- Scenario 2 ($S_1 = 145.00$): Assets = $\$422\text{k} \times 145 = 61.19\text{M JPY}$. Equity = $61.19\text{M} - 45.225\text{M} = 15.965\text{M JPY}$. P&L = $+0.965\text{M JPY} \Rightarrow \text{ROI} = +6.43\%$.
- Scenario 3 ($S_1 = 130.00$): Assets = $\$422\text{k} \times 130 = 54.86\text{M JPY}$. Equity = $54.86\text{M} - 45.225\text{M} = 9.635\text{M JPY}$. P&L = $-5.365\text{M JPY} \Rightarrow \text{ROI} = -35.77\%$.
- Scenario 4 ($S_1 = 105.00$): Assets = $\$422\text{k} \times 105 = 44.31\text{M JPY}$. Equity = $44.31\text{M} - 45.225\text{M} = -0.915\text{M JPY}$. P&L = $-15.915\text{M JPY} \Rightarrow \text{ROI} = -106.10\%$ (Insolvency / Deficit).

④ Wipeout Threshold: Terminal equity reaches exactly zero when

$S_1 = 45,225,000 / 422,000 = 107.16825 \approx 107.17 \text{ JPY per USD}$ (a 28.6% USD depreciation in JPY terms); a broker margin call triggers well before zero.

Appendix: Case Study: August 2024 JPY Carry Unwind

The Dynamics of the August 5, 2024 Crash:

- Pre-Conditions: For over two years, hedge funds and retail investors borrowed near-zero-interest yen to fund US equities, Big Tech, and high-yield EM currencies (MXN, BRL). Gross short JPY positions reached record highs on the CFTC IMM report.
- The Double Catalyst:
 - ① July 31, 2024: Bank of Japan Governor Ueda raised the policy target rate to 0.25% and announced a quantitative tapering plan.
 - ② August 2, 2024: US Non-Farm Payrolls printed significantly below expectations (114k vs 175k consensus), driving 10Y US Treasury yields down roughly 30 bps over two sessions (Aug 1–2).
- The Unwind: USD/JPY collapsed from 161.90 in early July to below 142.00 on August 5.
- Market Contagion: Japan's Nikkei 225 dropped 12.4% in a single day (worst drop since 1987) as funds liquidated domestic equities to cover yen-denominated margin calls.

(Aquilina, Lombardi, Schrimpf, and Sushko, BIS Bulletin No. 90 (27 August 2024))

Appendix: Portfolio Optimization with Risk Constraints

Analytical Solution to the Mean-Variance Currency Model: Objective: $\max_h h' \alpha - \frac{\lambda}{2} h' V h$ subject to $1' h = 0$.

Setting up the Lagrangian:

$$\mathcal{L}(h, \mu) = h' \alpha - \frac{\lambda}{2} h' V h - \mu(1' h)$$

First-Order Conditions with respect to h :

$$\frac{\partial \mathcal{L}}{\partial h} = \alpha - \lambda V h - \mu 1 = 0 \implies h^* = \frac{1}{\lambda} V^{-1} (\alpha - \mu 1)$$

Multiplying by $1'$ and imposing $1' h = 0$ solves for the multiplier μ :

$$\mu = \frac{1' V^{-1} \alpha}{1' V^{-1} 1}$$

Optimal Position: Each currency's allocation h_i^* depends on its relative alpha adjusted for covariance risk across the entire universe.

Frictionless Benchmark: This closed form solves the frictionless benchmark ($\theta = 0$). With proportional transaction costs $\theta \text{TC}(|\Delta h|)$, the objective is non-differentiable at $\Delta h = 0$, generating optimal no-trade regions that practitioners solve using numerical quadratic or conic programming.

Appendix: A Numerical Diversification Example

Consider three hypothetical strategies. Each has annual expected excess return 4%, annualized volatility 10%, and zero correlation with the other two. Invest one third in each.

$$E[R_p] = 4\%, \quad \sigma_p^2 = 3 \left(\frac{1}{3} \right)^2 (0.10)^2, \quad \sigma_p = 5.77\%.$$

	Single strategy	Equal-weight combination
Expected excess return	4.00%	4.00%
Volatility	10.00%	5.77%
Expected-return / volatility ratio	0.40	0.69

The improvement comes from the assumed correlations. If all three become perfectly correlated, portfolio volatility rises to 10%. Actual factor returns, correlations, costs, and drawdowns must be measured from the chosen sample.

Appendix: A Historical Test of Carry Timing

Melvin, Prins, and Shand (2013) study a carry index that exits when their financial-stress index exceeds 1.

2000–2008 sample	Static carry	Stress-conditioned carry
Cumulative return	−1%	+38%
Information ratio	−0.03	0.69
Information ratio, execution delayed one month	—	0.42

- These historical simulations omit transaction costs. The same-month rule assumes rapid recognition and execution.
- The deterioration with a one-month delay shows why implementation matters. The result does not guarantee protection in a future crash.

Source: Melvin, Prins, and Shand (2013), pp. 745–747, Figure 13.8.

Institutional Structure of the Hong Kong Currency Board:

① Four Components of the Monetary Base:

- Certificates of Indebtedness (backing legal tender banknotes issued by HSBC, StanChart, BOCHK).
- Government-issued coins and notes.
- Aggregate Balance: Reserve accounts held by commercial banks at the HKMA.
- Exchange Fund Bills and Notes (EFBNs).

② Convertibility Zone:

- Strong-Side: HKMA commits to buy USD at 7.7500.
- Weak-Side: HKMA commits to sell USD at 7.8500.

③ Carry Outflows and Adjustment: When HKD funding is cheaper than comparable USD funding, investors may borrow HKD and buy USD. Outflows can push spot to 7.85, triggering reserve absorption. The rate response depends on liquidity; unhedged carry is not an arbitrage.

Appendix: Example 3 Detailed HKMA T-Accounts

Complete Balance Sheet Ledger for Weak-Side CU Execution:

- ① Step 1: Bank Requests Outflow Conversion: Commercial Bank A delivers 7.85B HKD to HKMA and requests \$1.0B USD to remit overseas.
- ② Step 2: HKMA Settles via Exchange Fund:
 - HKMA transfers \$1.0B USD from its US Federal Reserve account to Bank A's overseas correspondent.
 - HKMA debits Bank A's reserve account in Hong Kong by 7.85B HKD, cancelling the liabilities.
- ③ Step 3: Interbank Market Transmission:
 - Total Aggregate Balance drops from 45B HKD → 37.15B HKD.
 - If reserves become scarce, banks bid up rates for settlement funds.
 - Higher HIBOR can narrow the gap against comparable-tenor USD rates.
- ④ Liquidity Buffer & Carry Risk:
 - In 2025, HKMA injected HK\$129.4bn in early May and absorbed HK\$119.9bn from late June to mid-August. The initial liquidity buffer matters for subsequent rates.
 - A persistent rate gap can motivate carry, but does not guarantee profit after FX risk and costs.

Appendix: Trilemma Trade-offs Across Global Economies

Regime Choice	Policy Pair Retained	Examples & Vulnerabilities
Side A: Floating	Capital Mobility + Monetary Autonomy	US, Euro Area, UK, Japan: High currency volatility; exports exposed to large spot swings.
Side B: Currency Board	Fixed FX Rate + Capital Mobility	Hong Kong: Total loss of domestic monetary policy; must import anchor-country rate hikes.
Side C: Capital Controls	Fixed FX Rate + Monetary Autonomy	China (Managed), India: Disconnects domestic rates from global markets; incurs evasion/efficiency costs.

Appendix: Central Bank Intervention T-Accounts

Defending the domestic currency: T-account comparison

1. Unsterilized FX Intervention:

Assets	Liabilities
Foreign Reserves: $-\$1\text{B USD}$	Domestic Bank Reserves: $-\$1\text{B equivalent}$
Net Change: Assets $-\$1\text{B}$	Net Change: M^s Falls by $\$1\text{B}$

Result: Bank reserves fall; rates can rise if liquidity becomes scarce.

2. Sterilized FX Intervention:

Assets	Liabilities
Foreign Reserves: $-\$1\text{B USD}$	Domestic Bank Reserves: $-\$1\text{B (FX Sale)}$
Domestic Treasury Bonds: $+\$1\text{B USD}$	Domestic Bank Reserves: $+\$1\text{B (Bond Purchase)}$
Net Change: FX $-\$1\text{B}$, Bonds $+\$1\text{B}$	Net Change: M^s Unchanged (0)

Result: Bank reserves are restored; signaling and portfolio effects can remain.

Appendix: Real Exchange Rates and the Terms of Trade

- The CPI real exchange rate is $q = SP_F/P_H$, with S in home currency per foreign unit. A rise means home real depreciation under this convention.
- The terms of trade compare export prices with import prices. They are distinct from a CPI real exchange rate.
- Greater relative demand for home goods can appreciate their relative price. Greater supply of a home export can lower its price in a simple trade model.
- Productivity growth does not have a universal exchange-rate sign. Higher tradable productivity can appreciate the CPI real rate through Balassa–Samuelson wages and nontradable prices, as in Week 2.

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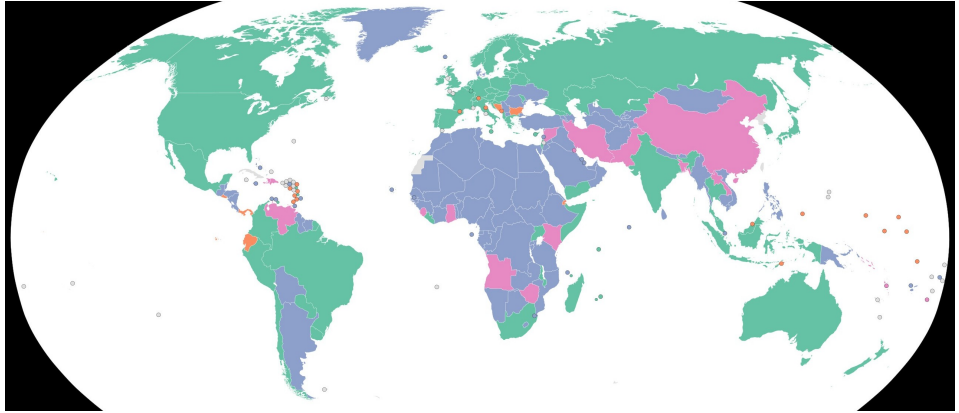
Appendix: A Nominal and Real Forecast Calculation

Let S be USD per GBP, with $S_0 = 1.20$ in Case B. All forecasts cover one year; i denotes nominal interest and inflation is deterministic.

Input	Case A	Case B
US / UK one-year inflation	3% / 2%	2% / 1%
Assumption	$q_1/q_0 = 1.02$ UIP, $i_{US} = 5\%$, $i_{UK} = 3\%$	

- Case A: $S_1/S_0 = (q_1/q_0)(1.03/1.02) = 1.03$. The real depreciation assumption gives 3% nominal depreciation of the USD in this convention.
- Case B: $E[S_1] = 1.20(1.05/1.03) = 1.22330$. With deterministic inflation, $E[q_1]/q_0 = (1.05/1.03)(1.01/1.02) = 1.00942$.
- Real interest rates satisfy $1 + r = (1 + i)/(1 + \pi)$ here: Case B gives $r_{US} = 2.94\%$ and $r_{UK} = 1.98\%$. UIP alone does not equalize real rates; relative PPP is an additional condition.

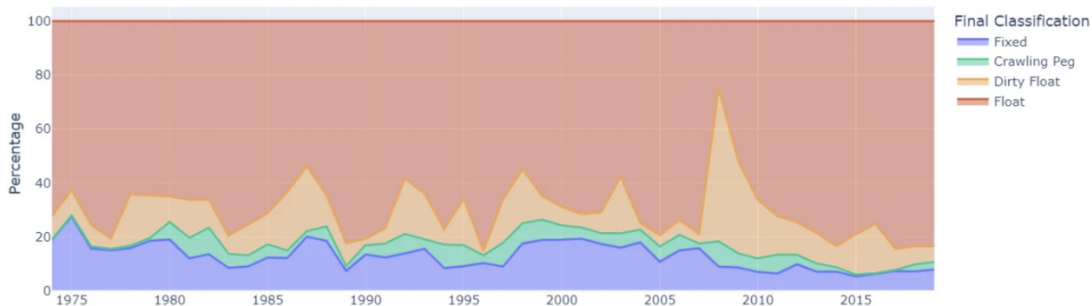
Global Exchange Rate Regimes (2023 IMF Report)



Historical exhibit. Data period, coverage, and units are as labeled.

Source: Wikipedia.

Post-BWS Exchange Rate Classification



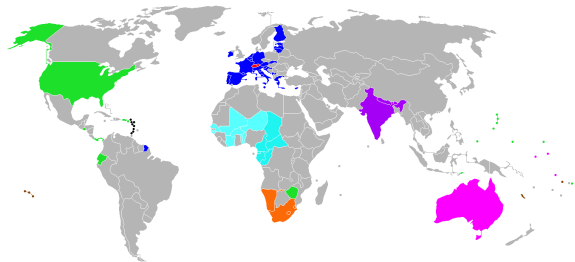
Historical exhibit. Data period, coverage, and units are as labeled.

Source: Levy-Yeyati & Sturzenegger (2024).

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Currency Use, Unions, and Pegs

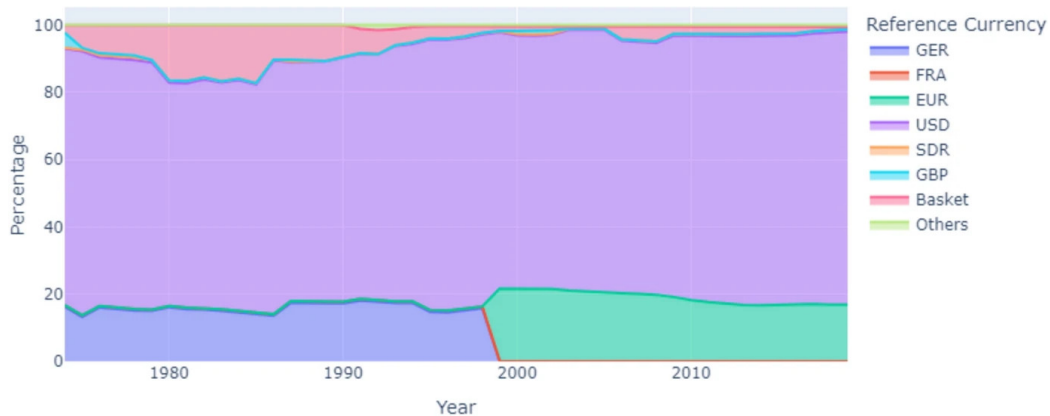
	EUR Euro
	USD United States dollar
	CHF Swiss franc
	INR Indian rupee
	AUD Australian dollar
	XCD Eastern Caribbean dollar
	ZAR South African rand
	XOF West African CFA franc
	XAF Central African CFA franc
	XPF CFP franc



Historical map of currency use and links. Currency adoption and bilateral pegs need not create a monetary union or a shared policy vote.

Source: Wikipedia.

Peg Reference Currency

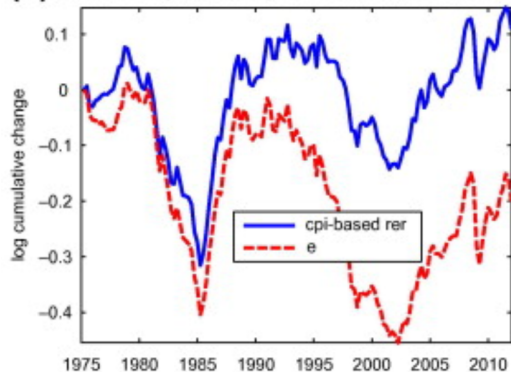


Historical exhibit. Data period, coverage, and units are as labeled.

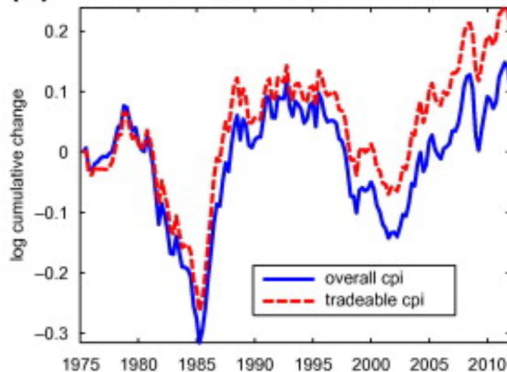
Source: Levy-Yeyati & Sturzenegger (2024).

Real vs Nominal Exchange Rates for the US

(a) US Trade Weighted Exchange Rate: Nominal and Real CPI-based



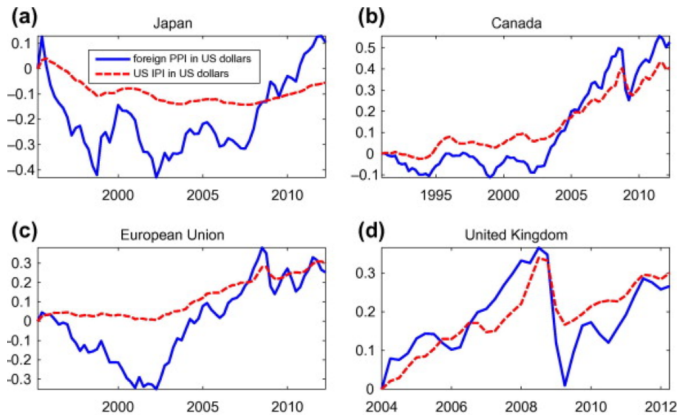
(b) US CPI-based RER: Overall and tradeable-based



Historical exhibit. Data period, coverage, and units are as labeled.

Source: Burstein & Gopinath (2014).

Real Exchange Rates (RER) Across Countries



Historical exhibit. Data period, coverage, and units are as labeled.

Source: Burstein & Gopinath (2014).